Control without Trust: A Distributionally Robust Approach

Fact Sheet

Project Information

TRUST

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31 July 2026

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EXCELLENT SCIENCE - European Research Council (ERC)

Total cost
€ 1 499 515,00

EU contribution
€ 1 499 515,00

Coordinated by
TECHNISCHE UNIVERSITEIT DELFT
Netherlands

Project description

No more "hindsight is 20/20" thanks to novel optimisation of decision-making

Decision-making in engineering as in real life must deal with inherent uncertainty arising from several sources, including access to limited observable data, measurement noise and so-called operator error. Numerous optimisation techniques exist to account for uncertainty with varying starting assumptions but, as the data sets become larger and more complex, so does the uncertainty. The EU-funded TRUST project is developing a novel control paradigm based on the emerging field of distributionally robust optimisation. It takes into account all probability distributions consistent with information previously provided to address the new challenges. The outcomes will include a new modelling language available as open-source software.

Objective

"Recent developments in sensing and communication technology offer unprecedented opportunities by ubiquitously collecting data at high detail and at large scale. Utilization of data at these scales, however, poses a major challenge for control systems, particularly in view of the additional inherent uncertainty that data-driven control signals introduce to systems behavior. In fact, this effect has not been well understood to this date, primarily due to the missing link between data analytics techniques in machine learning and the underlying physics of dynamical systems.

I address this issue by proposing a novel control design paradigm embracing ideas from the emerging field of distributionally robust optimization (DRO). DRO is a decision-making model whose solutions are optimized against all distributions consistent with given prior information. Recent breakthrough work, among others by the PI of this proposal, has shown that many DRO models can be solved in polynomial time even when the corresponding stochastic models are intractable. DRO models also offer a more realistic account of uncertainty and mitigate the infamous ""post-decision disappointment"" of stochastic models.

This proposal lays the theoretical foundation for distributionally robust control and aims to make progress along four directions. (i) Decision-dependent ambiguity: I introduce the concept of ""invariant ambiguity sets"" to encompass the dynamic evolution of uncertainty. (ii) Dynamic programming: I establish a dynamic programming characterization of the proposed DRO models and provide tractable approximation schemes along with rigorous theoretical bounds. (iii) Safe and memory-efficient learning: Leveraging modern tools from kernel methods and online-optimization, I propose tractable, yet provably reliable, synthesis tools. (iv) I plan to develop a tailor-made modeling language and open-source software to make distributionally robust control methods accessible in industry-size applications."

Fields of science (EuroSciVoc)

natural sciences > computer and information sciences > **software**

Keywords

Control Distributionally Robust Optimization Dynamic Programming

Convex Optimization Statistical Learning

Programme(s)

H2020-EU.1.1. - EXCELLENT SCIENCE - European Research Council (ERC) MAIN PROGRAMME

See all projects funded under this programme

Topic(s)

ERC-2020-STG - ERC STARTING GRANTS

See all projects funded under this topic

Funding Scheme

ERC-STG - Starting Grant

See all projects funded under this funding scheme

Call for proposal

[ERC-2020-STG](#)

See all projects funded under this call

Host institution

TECHNISCHE UNIVERSITEIT DELFT

Net EU contribution

€ 1 499 515,00

Total cost

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Netherlands

Activity type

Higher or Secondary Education Establishments

Links

Contact the organisation Website
Participation in EU R&I programmes
HORIZON collaboration network

Beneficiaries (1)

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